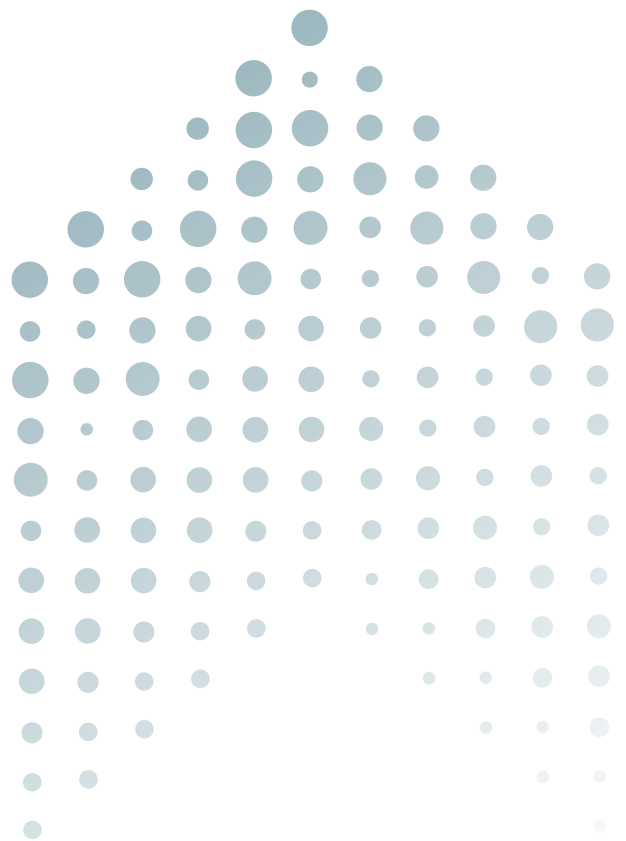


# Certification in Data Science & AI Session Plan

Duration: 300 Hours



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|--------|---------------------------------------------------------|--------------|
| 1      | Module 1: Excel-based EDA & Introductory Statistics     | 40           |
| 2      | Module 2: Python Programming for Data Science           | 50           |
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# Module 1: Excel-based EDA & Introductory Statistics (Prep Course – Non-Programming)

Duration: **40 Hours**

Objective: **Build foundation in data understanding and statistical thinking using Excel.**

## Understanding Data & Problem Statement

- What is data? Types (structured/unstructured)
- Understanding business problems and converting them to analytical problem statements
- Importance of data context in decision making

## Categorical & Numerical Data Analysis

- Univariate vs multivariate analysis
- Measures: mean, median, mode, percentiles
- 3M analysis: Mean, Median, Mode on numerical features

## Data Visualization in Excel

- Histograms, boxplots, bar charts
- Cross-tabulation & pivot tables
- Creating dashboards for summary views

## Descriptive & Inferential Statistics

- Central tendency, dispersion, skewness, kurtosis
- Normal distribution, Bernoulli, binomial distributions
- Covariance vs correlation, causation

## Hypothesis Testing

- Population vs sample, confidence intervals
- T-test, Chi-Square, ANOVA
- P-values and statistical significance

## Week 1

| Day   | Session    | Topic                                  | Description                                                                                                                   | Dur (hrs) |
|-------|------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 1  | Understanding Data & Problem Statement | Exploring types of variables and levels of measurement.                                                                       | 2         |
|       | Session 2  |                                        | Understanding what constitutes data and its different types: structured vs unstructured.                                      | 2         |
| Day 2 | Session 3  |                                        | Learning how to define a problem statement in analytics.                                                                      | 2         |
|       | Session 4  |                                        | Discussing real-world scenarios and mapping them to data problems.                                                            | 2         |
| Day 3 | Session 5  |                                        | Understanding business objectives and converting them into measurable metrics.                                                | 2         |
|       | Session 6  |                                        | Differentiating between nominal, ordinal, interval, and ratio scale variables                                                 | 2         |
| Day 4 | Session 7  | Categorical & Numerical Data Analysis  | Introduction to univariate analysis for categorical data using frequency tables.                                              | 2         |
|       | Session 8  |                                        | Analyzing numerical data using mean, median, and mode.                                                                        | 2         |
| Day 5 | Session 9  |                                        | Working with percentiles, ranges, and IQR for numerical summaries.                                                            | 2         |
|       | Session 10 |                                        | Identifying skewness and kurtosis to describe distribution shapes & Comparing univariate vs multivariate analysis techniques. | 2         |

## Week 2

| Day   | Session    | Topic                                | Description                                                                                                                   | Dur (hrs) |
|-------|------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 11 | Data Visualization in Excel          | Creating basic visualizations like histograms and bar charts in Excel.                                                        | 2         |
|       | Session 12 |                                      | Using boxplots and density plots to analyze distributions.                                                                    | 2         |
| Day 2 | Session 13 |                                      | Applying crosstab and pivot tables for categorical data exploration.                                                          | 2         |
|       | Session 14 |                                      | Building visual summaries using conditional formatting and charts & Creating simple dashboards for data insights.             | 2         |
| Day 3 | Session 15 | Descriptive & Inferential Statistics | Exploring measures of central tendency: mean, median, mode.                                                                   | 2         |
|       | Session 16 |                                      | Understanding dispersion: variance, standard deviation, and range.                                                            | 2         |
| Day 4 | Session 17 |                                      | Introduction to probability distributions: normal, binomial, log-normal.                                                      | 2         |
|       | Session 18 |                                      | Analyzing covariance and correlation and interpreting relationships & Explaining correlation vs causation with case examples. | 2         |
| Day 5 | Session 19 | Hypothesis Testing                   | Population and sampling techniques, Central Limit Theorem and its implications in hypothesis testing.                         | 2         |
|       | Session 20 |                                      | Performing t-tests for means and interpreting p-values, ANOVA and Chi-square tests                                            | 2         |

# Module 2: Python Programming for Data Science

Duration: **50 Hours**

Objective: **Equip students with Python programming and data wrangling skills.**

## Python Basics

- Setting up Anaconda, Jupyter, Google Colab
- Syntax, variables, data types, loops, control structures

## Functions & Data Structures

- User-defined functions, lambda expressions, scope
- Lists, sets, tuples, dictionaries, strings

## Modules, Exceptions, Debugging

- Importing modules, standard libraries (os, sys, math)
- Try/except blocks, debugging tips and tools

## NumPy & Pandas

- NumPy arrays, operations, broadcasting, slicing
- Pandas Series & DataFrames, loading/saving CSVs and Excel

## Data Manipulation

- groupby, merge, pivot, melt
- Handling missing data, filtering, sorting

## Basic Visualization

- Matplotlib: line, bar, scatter, histogram
- Seaborn: countplot, boxplot, heatmap, pairplot

### Week 3

| Day   | Session    | Topic                       | Description                                                                                                                             | Dur (hrs) |
|-------|------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 21 | Python Basics               | Setting up the Python environment using Anaconda and Jupyter Notebook.                                                                  | 2         |
|       | Session 22 |                             | Learning basic Python syntax, variables, and keywords.                                                                                  | 2         |
| Day 2 | Session 23 |                             | Understanding data types, arithmetic operations, and expressions, Implementing conditional statements (if-else) and logical operations. | 2         |
|       | Session 24 |                             | Setting up the Python environment using Anaconda and Jupyter Notebook, Working with syntax and keywords                                 | 2         |
| Day 3 | Session 25 | Functions & Data Structures | Creating custom functions and understanding scopes.                                                                                     | 2         |
|       | Session 26 |                             | Exploring Python lists: indexing, slicing, and operations.                                                                              | 2         |
| Day 4 | Session 27 |                             | Understanding dictionaries, tuples, and sets.                                                                                           | 2         |
|       | Session 28 |                             | Using string manipulation functions and loops on data structures.                                                                       | 2         |
| Day 5 | Session 29 |                             | Creating custom functions and understanding scopes.                                                                                     | 2         |
|       | Session 30 |                             | Understanding list operations and access methods                                                                                        | 2         |

### Week 4

| Day   | Session    | Topic                          | Description                                                                                                            | Dur (hrs) |
|-------|------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 31 | Modules, Exceptions, Debugging | Importing Python modules and exploring standard libraries.                                                             | 2         |
|       | Session 32 |                                | Understanding the purpose of packages and how to structure Python projects.                                            | 2         |
| Day 2 | Session 33 |                                | Using `sys` and `os` modules for system-level tasks, Handling errors using try-except blocks and debugging techniques. | 2         |
|       | Session 34 |                                | Importing modules and common Python tools, Understanding the purpose of packages and how to structure Python projects. | 2         |
| Day 3 | Session 35 | NumPy & Pandas                 | Creating and manipulating arrays using NumPy, Understanding vectorized operations and broadcasting.                    | 2         |
|       | Session 36 |                                | Reading and exploring data using Pandas DataFrames.                                                                    | 2         |
| Day 4 | Session 37 |                                | Applying filtering, selection, and aggregation on tabular data.                                                        | 2         |
|       | Session 38 |                                | Building and editing NumPy arrays.                                                                                     | 2         |
| Day 5 | Session 39 | Data Manipulation              | Grouping and summarizing data using groupby, Joining datasets using merge and concat functions.                        | 2         |
|       | Session 40 |                                | Pivoting and reshaping data using pivot and melt.                                                                      | 2         |

## Week 5

| Day   | Session    | Topic               | Description                                                                                 | Dur (hrs) |
|-------|------------|---------------------|---------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 41 | Data Manipulation   | Cleaning missing values and handling duplicates & Applying groupby for data aggregation     | 2         |
| Day 1 | Session 42 | Basic Visualization | Plotting data using Matplotlib: line, bar, and scatter plots.                               | 2         |
| Day 2 | Session 43 |                     | Using Seaborn for enhanced visualizations: heatmap, boxplot, pairplot.                      | 2         |
|       | Session 44 |                     | Customizing charts with labels, legends, and styles.                                        | 2         |
| Day 3 | Session 45 |                     | Creating multi-plot layouts for dashboards & Visualizing data using basic Matplotlib charts | 2         |

\*2nd session for Week 5 - Day 3 will start from next module, Refer module 3

# Module 3: Supervised Machine Learning (Track 1)

Duration: **60 Hours**

Objective: **Introduce ML concepts, regression, classification, evaluation.**

## ML Foundations & EDA

- Machine Learning vs AI vs Deep Learning
- Train-test split, data preprocessing recap
- EDA focused on predictive modeling

## Linear Regression

- Simple and multivariate regression
- Assumptions: Linearity, homoscedasticity, multicollinearity
- Residual analysis, interpreting coefficients

## Regularization Techniques

- Polynomial regression and overfitting
- Ridge and Lasso regression: bias-variance tradeoff
- Model selection using AIC, BIC

## Logistic Regression & Classification

- Binary classification using logistic regression
- Model evaluation: confusion matrix, ROC-AUC
- Precision, recall, F1 score, accuracy

## Week 5

| Day   | Session    | Topic                | Description                                                                                                                       | Dur (hrs) |
|-------|------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------|
| Day 3 | Session 46 | ML Foundations & EDA | Understanding types of ML models and typical workflow in real-world scenarios, Splitting datasets into training and testing data. | 2         |
| Day 4 | Session 47 |                      | Conducting exploratory analysis to identify data issues and relationships.                                                        | 2         |
|       | Session 48 |                      | Basics of encoding categorical variables and scaling numeric features.                                                            | 2         |
| Day 5 | Session 49 |                      | How supervised vs unsupervised learning apply in different use-cases.                                                             | 2         |
|       | Session 50 |                      | Stratified and time-aware splitting strategies for robust model evaluation.                                                       | 2         |

## Week 6

| Day   | Session    | Topic                     | Description                                                                                                                        | Dur (hrs) |
|-------|------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 51 | ML Foundations & EDA      | Using visualization to detect outliers and target imbalance, When to use label encoding vs one-hot encoding.                       | 2         |
|       | Session 52 |                           | Common ML pipeline stages from raw data to model evaluation, Avoiding data leakage during train-test split.                        | 2         |
| Day 2 | Session 53 | Linear Regression         | Linear regression fundamentals with single and multiple features, Checking linearity and equal variance assumptions in regression. | 2         |
|       | Session 54 |                           | Plotting and interpreting residuals, R-squared, and RMSE, Identifying and mitigating the effect of outliers.                       | 2         |
| Day 3 | Session 55 |                           | Multiple regression with feature interactions and transformations.                                                                 | 2         |
|       | Session 56 |                           | Using statistical tests to check regression assumptions.                                                                           | 2         |
| Day 4 | Session 57 |                           | Visual diagnostics with residuals and leverage plots.                                                                              | 2         |
|       | Session 58 |                           | Handling influence points using Cook distance.                                                                                     | 2         |
| Day 5 | Session 59 |                           | End-to-end regression workflow from EDA to evaluation, Dealing with violations of normality and homoscedasticity.                  | 2         |
| Day 5 | Session 60 | Regularization Techniques | What is regularization and when to use it in ML.                                                                                   | 2         |

## Week 7

| Day   | Session    | Topic                                | Description                                                                                                          | Dur (hrs) |
|-------|------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 61 | Regularization Techniques            | Comparing Ridge and Lasso regression outcomes.                                                                       | 2         |
|       | Session 62 |                                      | Cross-validation for tuning regularization strength.                                                                 | 2         |
| Day 2 | Session 63 |                                      | Explaining AIC, BIC, and adjusted R-squared in model selection.                                                      | 2         |
|       | Session 64 |                                      | Fitting Ridge and Lasso with scikit-learn, Visualizing regularization impact on coefficients.                        | 2         |
| Day 3 | Session 65 |                                      | Using grid search with CV for hyperparameter tuning.                                                                 | 2         |
|       | Session 66 |                                      | Model diagnostics using advanced performance metrics.                                                                | 2         |
| Day 4 | Session 67 |                                      | Use-case comparison of Ridge vs Lasso, Interpreting model complexity vs performance.                                 | 2         |
| Day 4 | Session 68 | Logistic Regression & Classification | Applying logistic regression for binary classification tasks, Understanding the sigmoid function and log-odds logic. | 2         |
| Day 5 | Session 69 |                                      | Measuring performance using confusion matrix and ROC & Precision, recall, and F1-score interpretation.               | 2         |
|       | Session 70 |                                      | Logistic regression pipeline for real-world data.                                                                    | 2         |

## Week 8

| Day   | Session    | Topic                                | Description                                                                                          | Dur (hrs) |
|-------|------------|--------------------------------------|------------------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 71 | Logistic Regression & Classification | Logistic regression pipeline for real-world data.                                                    | 2         |
|       | Session 72 |                                      | Decision thresholds and probability interpretation.                                                  | 2         |
| Day 2 | Session 73 |                                      | Explaining ROC-AUC and comparing classification models.                                              | 2         |
|       | Session 74 |                                      | Using classification report for performance summary.                                                 | 2         |
| Day 3 | Session 75 |                                      | Advanced logistic regression with interaction terms, Sigmoid curve visualization and cut-off tuning. | 2         |

\*2nd session for Week 8 – Day 3 will start from next module, Refer module 4

# Module 4: Feature Engineering & Model Tuning (Track IV)

Duration: **40 Hours**

Objective: **Deep dive into preprocessing, feature transformation, and model optimization.**

## Feature Engineering

- Handling categorical variables: label encoding, one-hot
- Missing value imputation: mean, median, mode, KNN
- Outlier treatment and feature scaling

## Dimensionality Reduction

- Feature selection using variance threshold, correlation matrix
- PCA (Principal Component Analysis) and its use cases

## Model Tuning

- What are hyperparameters? Impact on learning
- GridSearchCV and RandomizedSearchCV with examples
- Cross-validation techniques: K-Fold, Stratified K-Fold

## Model Selection

- Comparing models with metrics and complexity
- Bias-variance tradeoff visualization
- Learning curves and diagnostics

## Week 8

| Day   | Session    | Topic               | Description                                                                                                                                                              | Dur (hrs) |
|-------|------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Day 3 | Session 76 | Feature Engineering | Introduction to feature engineering and importance in model performance, Label encoding vs one-hot encoding with real examples.                                          | 2         |
| Day 4 | Session 77 |                     | Detecting and imputing missing values using mean, median, and KNN.                                                                                                       | 2         |
|       | Session 78 |                     | Outlier detection with boxplots, Z-score, and IQR.                                                                                                                       | 2         |
| Day 5 | Session 79 |                     | Feature transformation: log, binning, and polynomial features, Scaling data with MinMaxScaler and StandardScaler.                                                        | 2         |
|       | Session 80 |                     | Using visualization to detect outliers and target imbalance, When to use label encoding vs one-hot encoding & Feature construction: creating new features from raw data. | 2         |

## Week 9

| Day   | Session    | Topic                    | Description                                                                                                       | Dur (hrs) |
|-------|------------|--------------------------|-------------------------------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 81 | Dimensionality Reduction | Why reduce dimensions? Benefits and risks, Applying PCA and interpreting explained variance.                      | 2         |
|       | Session 82 |                          | Visualizing high-dimensional data using PCA/TSNE.                                                                 | 2         |
| Day 2 | Session 83 |                          | Combining feature selection with dimensionality reduction, Using correlation analysis to drop redundant features. | 2         |
|       | Session 84 |                          | Building low-dimensional pipelines for model input & Understanding hyperparameters and how they affect models.    | 2         |
| Day 3 | Session 85 | Model Tuning             | Using GridSearchCV with scoring and parameter grids.                                                              | 2         |
|       | Session 86 |                          | RandomizedSearchCV: faster tuning with randomness.                                                                | 2         |
| Day 4 | Session 87 |                          | Cross-validation strategies: k-fold, stratified, and time series, Hyperparameter tuning for regularized models.   | 2         |
|       | Session 88 |                          | Tracking tuning results and managing overfitting.                                                                 | 2         |
| Day 5 | Session 89 |                          | Evaluating models post-tuning for robustness.                                                                     | 2         |
| Day 5 | Session 90 | Model Selection          | Introduction to model selection and why it's crucial.                                                             | 2         |

## Week 10

| Day   | Session    | Topic           | Description                                                      | Dur (hrs) |
|-------|------------|-----------------|------------------------------------------------------------------|-----------|
| Day 1 | Session 91 | Model Selection | Comparing models with F1-score, accuracy, precision, and recall. | 2         |
|       | Session 92 |                 | Understanding ROC-AUC and confusion matrix.                      | 2         |
| Day 2 | Session 93 |                 | Bias-variance tradeoff and learning curves.                      | 2         |
|       | Session 94 |                 | Model complexity vs generalization performance.                  | 2         |
| Day 3 | Session 95 |                 | Choosing final model based on business and metric needs.         | 2         |

\*2nd session for Week 10 - Day 3 will start from next module, Refer module 5

# Module 5: Deep Learning (ANN, CNN, RNN, LSTM)

Duration: **70 Hours**

Objective: **Build and train deep learning models using Keras & TensorFlow.**

## ANN Basics

- Neuron model, perceptron, multilayer perceptron (MLP)
- Feedforward and backpropagation
- Activation functions: sigmoid, ReLU, tanh, softmax

## Keras & TensorFlow

- TensorFlow vs PyTorch overview
- Model creation in Keras: Sequential & Functional API
- Dropout, BatchNormalization, Optimizers

## Convolutional Neural Networks (CNNs)

- Convolution layer, filters, padding, strides
- Max-pooling, feature maps, RGB image processing
- CNNs in Keras: MNIST case study, Transfer Learning

## Recurrent Neural Networks (RNNs) & LSTMs

- Why RNNs? Sequential data processing
- LSTM/GRU vs simple RNNs
- Bidirectional RNNs, time series modeling

## Week 10

| Day   | Session     | Topic      | Description                                                                                  | Dur (hrs) |
|-------|-------------|------------|----------------------------------------------------------------------------------------------|-----------|
| Day 3 | Session 96  | ANN Basics | Understanding how artificial neurons mimic biological neurons.                               | 2         |
| Day 4 | Session 97  |            | Structure of perceptrons and multilayer perceptrons.                                         | 2         |
|       | Session 98  |            | Forward propagation and activation functions, Exploring activation: sigmoid, ReLU, and tanh. | 2         |
| Day 5 | Session 99  |            | Loss functions and how they measure model performance, Gradient descent and weight updates.  | 2         |
|       | Session 100 |            | Backpropagation: how errors are propagated, Building and training a basic neural network.    | 2         |

## Week 11

| Day   | Session     | Topic              | Description                                                                                          | Dur (hrs) |
|-------|-------------|--------------------|------------------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 101 | ANN Basics         | Overfitting and regularization in neural networks.                                                   | 2         |
|       | Session 102 |                    | Introduction to batch normalization, Applying dropout and its effect on training.                    | 2         |
| Day 2 | Session 103 |                    | Case study: classification with MLP using real-world data.                                           | 2         |
| Day 2 | Session 104 | Keras & TensorFlow | Getting started with Keras and TensorFlow.                                                           | 2         |
| Day 3 | Session 105 |                    | Keras Sequential and Functional APIs.                                                                | 2         |
|       | Session 106 |                    | Compiling models: loss functions and optimizers.                                                     | 2         |
| Day 4 | Session 107 |                    | Training and evaluating models in Keras, Model callbacks: EarlyStopping and ModelCheckpoint.         | 2         |
|       | Session 108 |                    | Saving and loading models, Custom layers and model debugging.                                        | 2         |
| Day 5 | Session 109 |                    | Monitoring with TensorBoard, Hyperparameter tuning in Keras, Transfer learning basics in TensorFlow. | 2         |
|       | Session 110 |                    | Hands-on mini-project using Keras and TensorFlow.                                                    | 2         |

## Week 12

| Day   | Session     | Topic                                | Description                                                                                | Dur (hrs) |
|-------|-------------|--------------------------------------|--------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 111 | Keras & TensorFlow                   | Why use CNNs for image data.                                                               | 2         |
| Day 1 | Session 112 | Convolutional Neural Networks (CNNs) | Convolution operations and feature detection, Understanding filters, padding, and strides. | 2         |
| Day 2 | Session 113 |                                      | Pooling layers and their purpose.                                                          | 2         |
|       | Session 114 |                                      | Building a CNN from scratch using Keras.                                                   | 2         |
| Day 3 | Session 115 |                                      | Activation functions in CNNs.                                                              | 2         |
|       | Session 116 |                                      | Data augmentation for better generalization.                                               | 2         |
| Day 4 | Session 117 |                                      | CNN regularization techniques.                                                             | 2         |
|       | Session 118 |                                      | CNN architecture: LeNet, AlexNet overview.                                                 | 2         |
| Day 5 | Session 119 |                                      | Implementing transfer learning in CNNs.                                                    | 2         |
|       | Session 120 |                                      | Using pretrained models: VGG16, ResNet & Visualizing CNN filters and feature maps.         | 2         |

## Week 13

| Day   | Session     | Topic                                    | Description                                               | Dur (hrs) |
|-------|-------------|------------------------------------------|-----------------------------------------------------------|-----------|
| Day 1 | Session 121 | Recurrent Neural Networks (RNNs) & LSTMs | Why standard neural networks fail on sequences.           | 2         |
|       | Session 122 |                                          | How RNNs work: architecture and flow.                     | 2         |
| Day 2 | Session 123 |                                          | Training RNNs and vanishing gradient problem.             | 2         |
|       | Session 124 |                                          | Introducing LSTM and GRU networks.                        | 2         |
| Day 3 | Session 125 |                                          | Architecture of LSTM cells.                               | 2         |
|       | Session 126 |                                          | Bidirectional RNNs and their advantages.                  | 2         |
| Day 4 | Session 127 |                                          | Stacked RNNs and time series modeling.                    | 2         |
|       | Session 128 |                                          | Embedding layers and sequence preprocessing.              | 2         |
| Day 5 | Session 129 |                                          | Sequence classification with LSTM in Keras.               | 2         |
|       | Session 130 |                                          | LSTM for text generation & Sentiment analysis using LSTM. | 2         |

# Module 6: Capstone Projects + Industry Exposure

Duration: **40 Hours**

Objective: **Apply all concepts to real-world problems and prepare for job market.**

## Capstone Planning

- Choosing the right problem statement
- Defining scope, timeline, dataset sources

## Implementation

- EDA, data preprocessing, model selection
- Training, evaluation, fine-tuning

## Presentation & Review

- Building a project report & visual dashboard
- Demoing models using Streamlit or notebooks
- Peer review and mentor feedback

## Career Readiness

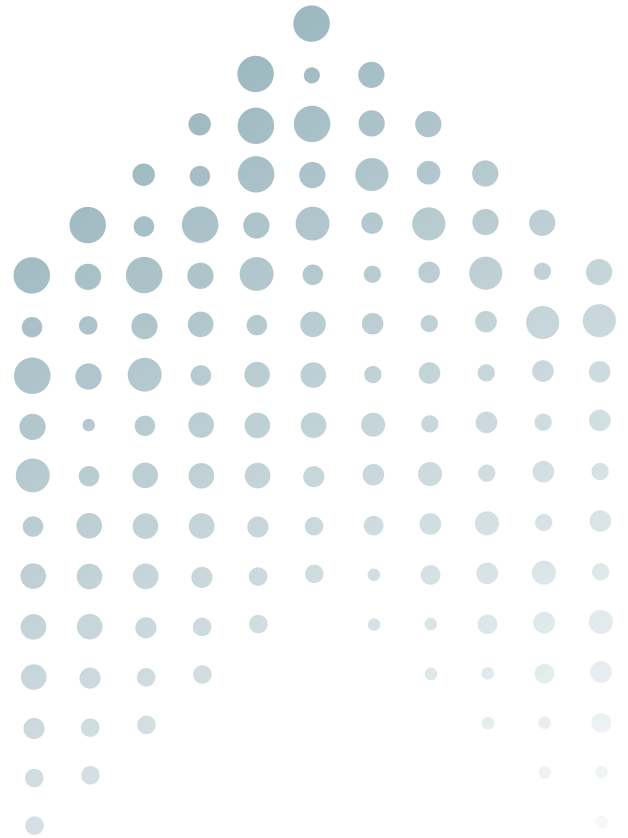
- Mock interviews, ML & Python questions
- Resume & LinkedIn profile building
- Guest sessions by industry experts

## Week 14

| Day   | Session     | Topic                 | Description                                                                                                        | Dur (hrs) |
|-------|-------------|-----------------------|--------------------------------------------------------------------------------------------------------------------|-----------|
| Day 1 | Session 131 | Capstone Planning     | Identifying a real-world problem suitable for a capstone project.                                                  | 2         |
|       | Session 132 |                       | Defining objectives, KPIs, and problem scope, Dataset sourcing and feasibility assessment.                         | 2         |
| Day 2 | Session 133 |                       | Outlining timelines, tools, and technology stack, Risk identification and mitigation planning.                     | 2         |
|       | Session 134 |                       | Structuring capstone proposal and milestone tracking & Performing EDA to derive initial insights from the dataset. | 2         |
| Day 3 | Session 135 | Implementation        | Cleaning and transforming data for modeling, Choosing appropriate ML/DL models for the problem.                    | 2         |
|       | Session 136 |                       | Training models and evaluating initial performance, Refining models through hyperparameter tuning.                 | 2         |
| Day 4 | Session 137 |                       | Ensembling models for improved accuracy.                                                                           | 2         |
|       | Session 138 |                       | Documenting modeling choices and performance metrics & Developing dashboards or interfaces for presentation.       | 2         |
| Day 5 | Session 139 | Presentation & Review | Conducting validation and peer review of results.                                                                  | 2         |
|       | Session 140 |                       | Creating a structured capstone report with visuals.                                                                | 2         |

## Week 15

| Day   | Session     | Topic                 | Description                                          | Dur (hrs) |
|-------|-------------|-----------------------|------------------------------------------------------|-----------|
| Day 1 | Session 141 | Presentation & Review | Storytelling with data and final recommendations.    | 2         |
|       | Session 142 |                       | Live demo preparation using notebooks or Streamlit.  | 2         |
| Day 2 | Session 143 |                       | Presenting to mentors or faculty for feedback.       | 2         |
|       | Session 144 |                       | Final project polish: packaging code and results.    | 2         |
| Day 3 | Session 145 | Career Readiness      | Scoring, evaluation, and showcasing best projects.   | 2         |
|       | Session 146 |                       | Resume building for Data Science and AI careers.     | 2         |
| Day 4 | Session 147 |                       | Preparing a strong LinkedIn and GitHub profile.      | 2         |
|       | Session 148 |                       | Crafting your capstone story for interviews.         | 2         |
| Day 5 | Session 149 |                       | Mock interviews: technical and behavioral.           | 2         |
|       | Session 150 |                       | Networking tips and how to approach hiring managers. | 2         |



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